

DevOps and Cloud Computing Foundations (CSC-)
Credit Hrs. 3 [Pre-Req : Computer Networks]
Google Classroom Code: 6jy2ghe6
Contact Hours: TBA

Semester: **Spring 2026**
Instructor: **Muhammad Ali**
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Course Objective:

This course provides an in-depth understanding of cloud computing concepts, architectures, and applications. It covers topics such as cloud delivery models (IaaS, PaaS, SaaS), virtualization, containerization, Kubernetes, cloud storage, and cloud security. The course combines theoretical knowledge with practical exercises using popular cloud platforms (Google Cloud Platform, AWS, Microsoft Azure).

Recommended Books: Soft Copies available

- A. *The DevOps Handbook* by Gene Kim, Jez Humble, Patrick Debois, and John Willis
- B. *Cloud Native DevOps with Kubernetes* by John Arundel and Justin Domingus

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- PLO5: Modern Tool Usage
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

Course Learning Outcomes (CLOs):

- CLO1: **Understand and explain** the core principles, tools, and practices of DevOps and cloud computing, including key concepts like IaaS, PaaS, SaaS, and virtualization technologies..
- CLO2: **Apply** cloud computing platforms (e.g., AWS, Azure, GCP) and tools (e.g., Docker, Kubernetes) to deploy and manage applications and infrastructure.
- CLO3: **Develop/Construct** CI/CD pipelines using DevOps tools and cloud services, focusing on automation, containerization, and orchestration.
- CLO4: Evaluate and integrate security practices into the DevOps lifecycle, ensuring secure deployments and maintaining best practices for cloud infrastructure and applications.

	PLOs
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Mapping of CLOs on PLOs		PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLOs	CLO1		✓								
	CLO2		✓								
	CLO3				✓						
	CLO4				✓						

Course Outline and Schedule:

Week	Topics	Map with CLOs	Map with PLOs	Text Book (Ref A)	Assessments
Wk-1 22-Sep	Introduction to DevOps and Cloud Computing: Core concepts, culture, and principles. The DevOps lifecycle.	CLO1	PLO1		
Wk-2 29-Sep	Version Control and Collaboration: Git fundamentals, branching strategies, and collaborative workflows.	CLO2	PLO2		
Wk-3 6-Oct	CI/CD with GitHub Actions: Setting up workflows, using runners, and automating builds.	CLO2	PLO2		Assignment 1
Wk-4 13-Oct	Containerization with Docker: Understanding containers, creating Dockerfiles, and managing images.	CLO2, CLO3	PLO2		Class Test 1
Wk-5 20-Oct	Orchestration with Docker Compose & Kubernetes: Deploying multi-container apps and an introduction to Kubernetes.	CLO2 CLO3	PLO5		Assignment 2
Wk-6 27-Oct	Introduction to Cloud Platforms: AWS, Azure,	CLO1	PLO2		Class Test 2

	GCP overview. Shared Responsibility Model.				
Wk-7 3-Nov	Infrastructure as Code (IaC) with Terraform: Writing and executing Terraform scripts to provision resources.	CLO2	PLO4		
Wk-8 10-Nov	Monitoring and Logging: Implementing monitoring tools (Prometheus, Grafana) and centralized logging.	CLO3	PLO3		Assignment 3
Wk-9 17-21 Nov	Midterm Exam				Mid Term 1
Wk-10 24-Nov	Security in DevOps: DevSecOps principles, integrating security into the pipeline.	CLO4	PLO8		
Wk-11 1-Dec	Serverless Computing: Concepts of FaaS, using AWS Lambda or Azure Functions.	CLO3	PLO4		Class Test 4
Wk-12 8-Dec	Final Project Introduction & Planning: Defining project scope, forming teams, and planning the pipeline.	CLO3	PLO6		Assignment 4
Wk-13 15-Dec	Project Implementation & Supervision: Hands-on work on the final project.	CLO3	PLO5		
Wk-14 22-Dec	Final Project Presentations: Team presentations of their complete DevOps pipeline.	CLO3	PLO6		Presentation Topic Assigned
Wk-15 29-Dec	Course Review and Future Directions: Recap of key concepts and emerging trends.	CLO4	PLO10		
Wk-16 5-Jan	Last Week of Classes, Final Project Presentations, Review of Key Learning Points, Future Directions	CLO3	PLO3		
Wk-17 10-13 Jan	Reinforcement / Preparation				

Wk-18 14-24 Jan	Final Exam Week				
29-Jan	Exam Scripts Shown				Need to confirm
30-Jan	Declaration of Result				

Assessment Description	Weight (%)	CLOs	PLOs
Quizzes (at least 3)	10%	CLO1, CLO2	PLO1, PLO2
Assignments (at least 3)	10%	CLO1, CLO2	PLO1, PLO2
Project	15%	CLO3	PLO4, PLO5, PLO7
Mid Term Exams	25%	CLO1, CLO2, CLO3	PLO2, PLO3, PLO4
Final Exam	40%	CLO1, CLO2, CLO3	PLO2, PLO3, PLO4

GRADING TABLE ABSOLUTE SCALE if no. of students in Final Exam are up to 20, otherwise RELATIVE SCALE			
Range	Percentage	Points	Grade
85.00-100	85.00	4.00	A
81.50-84.99	81.50	3.67	A–
78.00-81.49	78.00	3.33	B+
74.50-77.99	74.50	3.00	B
71.00-74.49	71.00	2.67	B–
67.50-70.99	67.50	2.33	C+
64.00-67.49	64.00	2.00	C
60.50-63.99	60.50	1.67	C–
57.00-60.49	57.00	1.33	D+
50.00-56.99	50.00	1.00	D
0.00-49.99	Below 50	0.00	F